2 Excess & Obsolete Inventory Optimization

2.1 Notation

The notation for the excess & obsolete (E&O) inventory optimization problem is summarized in Table 2.

Table 2: Notation for the E&O inventory optimization problem.

Indices and sets	
$(i, j \in)\ I$	Items
$I^{BW} \subset I$	Raw materials (items ending in "-BW")
$I^{FG} \subset I$	Finished goods (items ending in "-FG")
Parameters	
$\beta_{i,j} \geq 0$	BOM structures: $\beta_{i,j}$ units of raw material $i \in I^{BW}$ are used to produce one unit of finished good $j \in I^{FG}$ (if not given, then default: 0)
$S_i \geq 0$	Maximum available supply (in units) of raw material $i \in I^{BW}$ (default: ∞)
$D_i \geq 0$	Maximum available demand (in units) of finished good $i \in I^{FG}$ (default: ∞)
$Inv_{i,0} \geq 0$	Initial inventory (in units) of raw material $i \in I^{BW}$ (if not given, then default: 0)
$C_i \geq 0$	Per-unit cost of sourcing raw material $i \in I^{BW}$ (if not given, then default: 0)
$B \geq 0$	Maximum raw material investment (budget, must be given as main hyper-parameter)
Variables	
Continuous	
$Inv_{i,1}$	Inventory of raw material $i \in I^{BW}$ after production/depletion
x_i	Units of raw material $i \in I^{BW}$ sourced
y_i	Units of finished good $i \in I^{FG}$ produced

2.2 Problem Formulation #1

For this variation of the problem, we define B (likle in Table 2) as the maximum raw material investment (budget). This problem solves the following:

What is the minimum E&O inventory value Flex can achieve assuming the company can invest up to B in additional raw materials?

We wish to solve the following Mixed-Integer Linear Program (MILP):

Objective:

We wish to minimize the E&O inventory value after production/depletion:

$$(P_1(B)) \qquad \min_{\mathbf{Inv}_{\cdot,\mathbf{1}},\mathbf{x},\mathbf{y}} \sum_{i \in I^{BW}} C_i \; Inv_{i,1} \tag{21}$$

Constraints:

The inventory after production/depletion is equal to the initial inventory minus production:

$$Inv_{i,1} = Inv_{i,0} + x_i - \sum_{\substack{j \in I^{FG}: \\ \beta_{i,j} > 0}} \beta_{i,j} y_j, \qquad i \in I^{BW}, \tag{22}$$

Sourcing of raw materials (supplies) is bounded above:

$$x_i \leq S_i, \qquad i \in I^{BW}, \tag{23}$$

Production of finished goods (demand) is bounded above:

$$y_i \leq D_i, \qquad i \in I^{FG}, \tag{24}$$

Investments in additional raw materials are bounded above:

$$\sum_{i \in I^{BW}} C_i x_i \leq B. \tag{25}$$

2.3 Problem Formulation #2

We studied a variation of Problem Formulation #1 (which is not identical, but very similar). For this variation, we re-define B as the maximum E&O inventory value. This problem solves the following:

What is the minimum investment in additional raw materials Flex can make in order to lower its E&O inventory value to B or below?

We wish to solve the following MILP:

Objective:

We wish to minimize the investment in additional raw materials:

$$(P_2(B)) \qquad \min_{\mathbf{Inv}_{\cdot,\mathbf{1}},\mathbf{x},\mathbf{y}} \sum_{i \in I^{BW}} C_i \, x_i \tag{26}$$

Constraints:

The inventory after production/depletion is equal to the initial inventory minus production:

$$Inv_{i,1} = Inv_{i,0} + x_i + y_i - g_i - \sum_{\substack{j \in I^{FG} \cup I^{SFG}: \\ \beta_{i,j} > 0}} \beta_{i,j} y_j, \qquad i \in I^{BW}, \tag{27}$$

Sourcing of raw materials (supplies) is bounded above:

$$x_i \leq S_i, \qquad i \in I^{BW}, \tag{28}$$

Production of finished goods (demand) is bounded above:

$$y_i \leq D_i, \qquad i \in I^{FG}, \tag{29}$$

E&O inventory value is bounded below:

$$\sum_{i \in I^{BW}} C_i Inv_{i,1} \leq B. \tag{30}$$